1. Title Page

Lab Name: Lab 10: Control Plane Intro & OSPF Routing **Date:** YYYY-MM-DD **Name:** [Your Name] **Lab Partners:** [Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling Labs (CML) - Personal / Free Edition

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2. Background

The control plane in networking is responsible for determining the paths that data packets should take through a network, effectively building the map that the data plane uses for forwarding. Open Shortest Path First (OSPF) is a widely used interior gateway routing protocol that operates within the control plane, using a link-state algorithm to calculate the shortest path to all known network destinations. In this lab, you will be introduced to control plane concepts by configuring OSPF in your CML topology, observing how routers exchange link-state information to dynamically build and update their routing tables.

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3. Objective / Purpose

To configure Single-Area OSPF routing across a multi-router topology and verify dynamic route learning and failover redundancy.

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4. Topology Diagram

```
graph TD
    R1["Router R1<br>Area 0"] <--> |"10.0.12.0/24"| R2["Router R2<br>Area 0"]
    R1 <--> |"10.0.14.0/24"| R4["Router R4<br>Area 0"]
    R2 <--> |"10.0.23.0/24"| R3["Router R3<br>Area 0"]
    R4 <--> |"10.0.34.0/24"| R3

    classDef router fill:#fff2cc,stroke:#d6b656,stroke-width:2px;
    class R1,R2,R3,R4 router;
```

Details:

- 4x Routers in a redundant square or diamond topology.

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5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 4x Cisco IOSv Routers

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6. Configuration / Methodology

IP Addressing Table

Device	Interface	IP Address	Subnet Mask	OSPF Area

R1	G0/0	10.0.12.1	255.255.255.0	0
R2	G0/0	10.0.12.2	255.255.255.0	0
...	...	...	...	0

Key Configuration Snippets

```
! OSPF Configuration on R1
router ospf 1
 network 10.0.12.0 0.0.0.255 area 0
 network 10.0.14.0 0.0.0.255 area 0
```

Narrative

Instead of static routes, we use OSPF, a link-state routing protocol. Once configured on all routers, they will form adjacencies and exchange LSAs. We will then simulate a link failure to observe OSPF's convergence.

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7. Verification & Testing

1. Output of `show ip ospf neighbor`.
2. Output of `show ip route ospf`.
3. Traceroute before and after shutting down a primary link to demonstrate failover.

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8. Troubleshooting

Issue Encountered: [Describe what went wrong, if anything] **Discovery Method:** [How did you find the issue?] **Resolution:** [How did you fix it?]

(If no issues were encountered, explain potential pitfalls for this configuration).

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9. Conclusion & Analysis

Dynamic routing with OSPF successfully discovered the network topology and automatically recalculated routes when a link failure occurred.

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10. What to Hand In

- The Lab YAML file with your name, date, name of the lab at the top of the file in comments.
- A PDF with annotated screen shots showing the lab running, as well as screen shots of Wireshark captures of the lab showing OSPF routing protocol traffic (Hellos, LSAs).